

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Python Developer

Python Developer with experience designing and deploying production-grade computer vision systems for industrial applications for clients such as British Petroleum and Austin Industries. Hands-on experience with Python, Linux, Pandas, SciPy, TensorFlow, Docker, Git, and Airflow, with proven delivery of robust, scalable pipelines handling 3.6 million records. MSc Robotics graduate from University of Bristol with a strong focus on reliable Python solutions for research and development.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the full computer vision lifecycle for perception systems, from data curation to deployment and field maintenance using PyTorch and OpenVINO and improved F1 by 15% with deterministic 24/7 operation.
- Orchestrated a sensor-data cleansing pipeline for 3.6 million records using Pandas and Apache Airflow, reducing model rollout time from 3 days to under 24 hours.
- Built end-to-end data ingestion, cleansing, training, and validation pipelines for multimodal sensor datasets using MLflow, DVC, Optuna and open-source CVAT cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, improving fault isolation, increasing MTBF by 35%, and strengthening service continuity through logging, health checks, retries, and safe restarts.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, improving maintainability and reducing defects.
- Built synthetic image generation and augmentation procedures with OpenCV and SciPy to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Developed an internal web dashboard in Django to surface inference outputs, system health, and performance metrics, enabling faster debugging and root-cause analysis across deployed systems.

LivSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy end-to-end production AI models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices using ONNX and OpenVINO.
- Deployed quantised vision models to rugged Intel VPU edge devices using OpenVINO, delivering deterministic real-time inference across RGB and IR sensor streams in harsh industrial environments.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using OpenCV and Albumentations.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Mentored engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.
- Implemented optical flow–based ego-motion estimation feeding motion representations into CNNs to classify vehicle state in production pipelines.
- Built asynchronous inference pipelines using OpenVINO and C++17 to maximise throughput on Intel VPU edge hardware under real-time latency and power constraints.

- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system using RTAB-Map, integrating multimodal LiDAR and radiological data for localisation and mapping in inspection scenarios leveraging the Clearpath Husky development platform.
- Evaluated RTAB-Map (LiDAR-only), LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation prior to real world deployment.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

SKILLS

- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Intel VTune, Linux, NumPy, Pandas, PG vector, PostgreSQL, Python, Qt, SciPy
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow
- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, Layer fusion, MultiRocket, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, Vision Transformer (ViT), YOLO
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership